

Team Rho Final Written Deliverables Report:

Predicting Future State-Level Psychiatric Inpatient Demand Using Suicide- and Crisis-Related Google Trends Indicators

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NOTE: Each author contributed equally to the design, coding & development, analysis, and writing of this project

GitHub Repository:

<https://github.com/brendancollari/Team-Rho-Capstone-Project>

Synopsis

The focus of team Rho's current project investigates whether suicide and crisis related google search activity can effectively predict future psychiatric inpatient admissions at the state level. Data was collected from Google Trends, HCUP, and the U.S. Census Bureau from 2014-2023. Data analysis showed that historical admission trends and geographic patterns were much stronger predictors than google search activity. Our baseline model explained approximately 83% of the variation in unseen test data. Advanced model development revealed that historical utilization was the strongest predictor of future admissions, with geographic relationships also providing predictive value. Within the context of this study, Google Trends did not demonstrate sufficient predictive value to serve as standalone predictors of future inpatient demand.

Introduction

Background

Mental healthcare within the United States is under strain and overburdened. Workforce shortages and resource allocation are big issues in the mental health field. Because of this psychiatric inpatient facilities are under lots of stress. Care has started to switch to outpatient models due to inpatient facilities becoming less and less available for those that need it. This is particularly important because psychiatric inpatient capacity remains constrained, with demand exceeding available beds in many areas and patients sometimes waiting in emergency departments for inpatient care (Saunders & Euhus, 2024). It is happening all over the United States with suicide and crisis related mental health emergencies continuing to grow throughout the past 10 years. It has become a significant public health issues with nearly 50,000 deaths from suicide and mental health crises in 2024 (CDC, 2026).

With the strain on mental healthcare, there becomes less and less opportunity for those that need care to get their needs met and when the system fails or someone's healthcare needs are not fulfilled, the burden does not disappear but the weight shifts to places like correctional facilities or emergency rooms which also experience similar overload. This becomes a huge issue as these facilities are not as equipped to give the necessary mental healthcare and psychiatric needs to those that end up there (Alabi et al., 2026). And beyond the toll it takes at the human level, inadequate mental healthcare and resources come with financial costs like the direct cost of hospitalization and more broad economic consequences of inaccessible or lack of care needs (Tibirna et al., 2026).

Mental health systems need better ways to anticipate demand so that limited psychiatric resources can be optimized before patients need them. To combat this rise in healthcare needs and an overworked healthcare system, an approach of trying to anticipate public health needs is emerging. One of the ways to do this is using search behavior as a tool for surveillance of the public's behavior and opinions. Google Trends data reflects real time public search interest and can be used to monitor lots of different trends within the public. At the same time, researchers have found that Google search activity can provide timely information about population mental-health concerns, although its usefulness varies by topic and should be interpreted cautiously (Vaidyanathan et al., 2022). These findings suggest that combining publicly available search behavior with historical admission and geographic information may provide a valuable source of information for anticipating future psychiatric service demand.

For this project the Google trends data will be used to try and predict the next year's psychiatric admissions for each state within the United States. The project builds on prior work but focuses specifically on suicide and crisis related search terms rather than broader, more generalized

mental health search terms and links the suicide and crisis related terms to state level psychiatric admissions healthcare data. Which will help create a better understanding for future inpatient admissions and allow for states to create actionable plans for healthcare resource allocation and workforce scheduling. This project also directly tests whether search based indicators meaningfully improve prediction beyond what historical and demographic data alone can offer.

Question

How well can state-level psychiatric admission rates be predicted using suicide- and crisis-related Google Trends data, geographic information, and historical admission patterns?

Hypothesis & Prediction

Hypothesis: An increase in suicide- and crisis-related Google search activity reflects increasing mental-health concern and will be associated with higher future state-level psychiatric admission rates.

Prediction: Combining Google Trends indicators with historical admission rates, demographic information, geographic information, and lagged predictors will improve forecasting performance compared with models relying only on historical utilization and demographic information.

Data

Data Acquisition

Team Rho used three primary data sources: Google Trends, HCUP Fast Stats, and U.S. Census Bureau population estimates. These sources were selected because they provide complementary information needed to answer the research question: Google Trends provides a measure of public search behavior, HCUP provides observed psychiatric inpatient utilization, and Census data provide population estimates needed to make useful state-to-state comparisons.

The final analytical dataset contained 452 state-year observations representing 46 states from 2014–2023. The target variable was psychiatric admission rate per 100,000 residents, calculated as:

$$\text{Psychiatric Admission Rate} = (\text{Annual Psychiatric Admissions} / \text{State Population}) \times 100,000$$

The Google Trends predictors represented eight suicide-, crisis-, and mental-health-related searches. Annual search interest from the previous year was used to predict psychiatric admissions in the following year. This one-year lag matches the project's goal of evaluating Google Trends as a potential future indicator rather than simply measuring conditions occurring at the same time as admissions.

Team Rho recognized that a key data-collection limitation is that Google Trends reports relative search interest rather than absolute search volume. Simply stated, search behavior may differ across states because of differences in population, internet access, and how people use search engines, rather than reflecting absolute need. Similarly, HCUP admissions measure observed

healthcare utilization, which can reflect patient need, but may also be distorted by the limitations of local psychiatric services.

Cleaning

The three datasets were first inspected for variable names, data types, state identifiers, years, and duplicate observations. Google Trends monthly observations were aggregated to annual state-level values. HCUP quarterly psychiatric admission records were aggregated to annual totals. Census population estimates were then matched to state and year and used to calculate admission rates per 100,000 residents.

The datasets were joined using state and year as the common identifiers. One-year lagged Google Trends variables were created by shifting each state's annual search activity forward one year so that searches from year t were used to predict admissions in year $t + 1$. This preserves the intended forecasting direction and avoids using future information to predict the past (averting temporal leakage).

The final dataset contained no duplicate state-year observations. Two observations contained missing lagged Google Trends values because prior-year search information was unavailable. These observations were retained during EDA and addressed during later modeling as necessary. No extensive imputation or variable removal was performed during EDA in order that potentially useful predictors would not be removed before model evaluation.

Final Dataset EDA

Since new project input required that additional variables be added following the initial exploratory data analysis (EDA), a second EDA was performed on the latest comprehensive dataset. The newly engineered dataset contains 452 state-year observations across 46 states from 2014–2023. It combines lagged Google Trends search activity with two new predictors: prior psychiatric admission rates and neighboring-state admission rates.

The latest EDA does not show a strong simple relationship between Google search activity and future psychiatric admissions. Most Google Trends variables have weak correlations with the outcome (approximately -0.16 to $+0.05$). This means that higher search activity, by itself, does not appear to consistently predict future admission rates.

The strongest relationship with future admissions is the temporal predictor (admission rate from 2 years previous), with a strong correlation of $r = 0.93$. Given this, and the fact that neighboring state admission rates correlate at $r = 0.35$ with future admissions, it is implied that historical and geographic patterns may be more predictive of future admissions than google search activity alone.

Among Google Trends variables, ‘mental health crisis’ and ‘crisis hotline’ show the strongest relationships, both at approximately $r = -0.16$, followed by ‘suicide prevention’ ($r = -0.12$) and ‘suicide hotline’ ($r = -0.11$). These weak correlations do not provide evidence that search activity reliably predicts admissions.

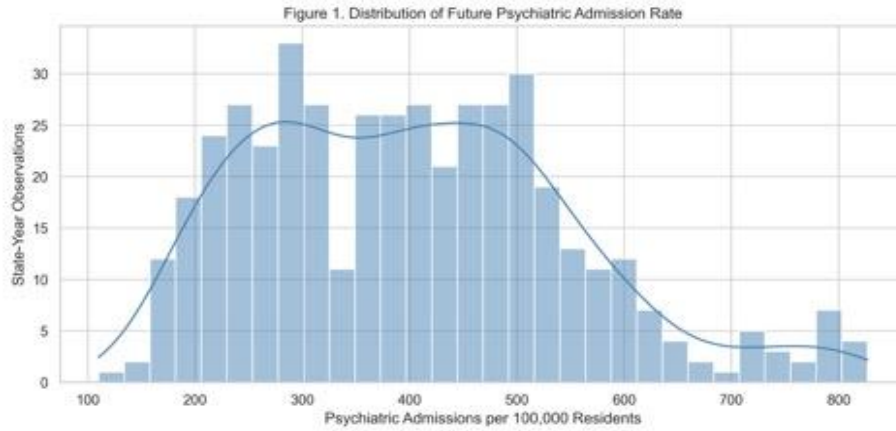


Figure 1: Distribution of Future Psychiatric Admission Rate.

The outcome varies substantially across state-year observations, indicating differences in psychiatric admission rates that may be explored during the modeling stage.

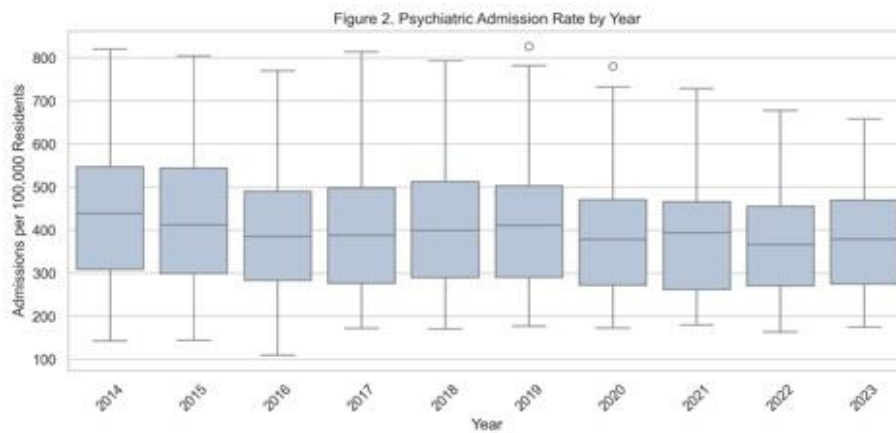


Figure 2: Psychiatric Admission Rate by Year.



Figure 3: Google Trends Search Interest vs. Future Psychiatric Admission Rate.

The scatterplots show generally weak relationships between individual Google Trends measures and future admissions. Google Trends may best be considered as part of a multivariable predictive model rather than treated as a standalone predictor during final modeling.

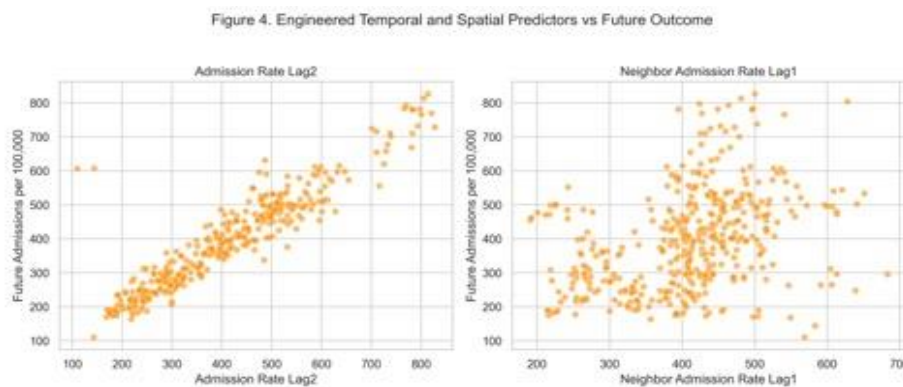


Figure 4: Engineered Temporal and Spatial Predictors vs. Future Outcome.

The two engineered variables show greater promise. Prior admission rates show a particularly strong relationship with future admissions, while neighboring-state rates provide useful spatial information.

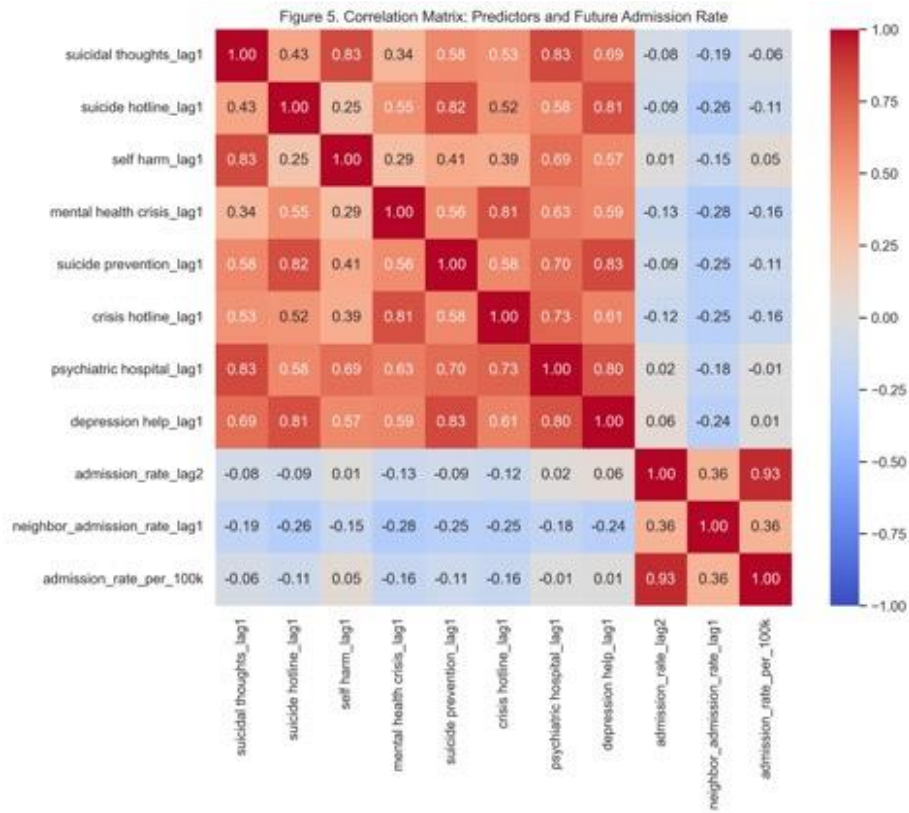


Figure 5: Predictor Correlation Matrix.

As in the first EDA, Google Trends variables are also substantially correlated with one another, indicating potential multicollinearity. This will be addressed during preprocessing.

Data-Readiness

There are no duplicate rows, but missingness requires treatment before modeling: each Google Trends variable has 2 missing observations, neighboring-state admission rate has 44, and the two-year admission-rate lag has 92. No imputation was performed during EDA.

Given the latest EDA findings, it is useful to change the central question from “*Do Google searches predict admissions?*” to “*Do Google Trends provide predictive information beyond strong historical and spatial admission patterns?*”

Preprocessing & Feature Engineering

Train-Test Split

The data were divided chronologically rather than randomly so that the model would always be trained on earlier years and evaluated on later years. A 70:30 train-test split was selected, resulting in training data from 2014–2020 and testing data from 2021–2023. This approach was selected because the project is designed to predict admission rates, and random splitting could result in temporal leakage (future data influencing results). This ratio was chosen because a smaller dataset requires a larger proportion of data for the training. While optimal split ratio theory can help guide this kind of decision, it does not translate directly to a temporal split here. The test period does contain pandemic years which might affect how well the test set performance generalizes to the non pandemic years in the training set.

The one-year lag on Google Trends variables was retained so that search activity from year t was used to predict psychiatric admissions in year $t + 1$. The one year interval was used instead of quarterly because many states reported the HCUP admissions data inconsistently at the quarterly timeframe. Also aggregating to annual totals was necessary to save state coverage across the panel, a quarterly lag would've created lots of missing data points and increased inconsistency. A

two-year admission-rate lag and a one-year neighboring-state admission-rate lag were also created to provide additional historical and geographic information.

Data Cleaning

Two observations were removed because they did not have the necessary prior-year information to calculate the required lagged predictors. The missing values arose from the construction of the one year search term lag rather than from the missing measurements. A larger reduction also occurred from the two year admission rate lag as it required two years of prior history data rather than just one. No imputation was performed because the missing values were limited to observations where the required historical information did not exist.

The `mental_health_admissions` variable was removed before modeling because it was used to calculate the target admission rate. Retaining it would allow the model to indirectly see the outcome it was being asked to predict and would likely create data leakage.

Scaling and Transformation

The eight Google Trends variables were standardized using the training data only.

Standardization placed the search variables on a comparable scale to prevent variables with larger variation from skewing results.

State population was log-transformed because the EDA showed substantial right skew. The transformation reduced the population skew from 2.44 to approximately -0.15 . This remaining skew value is on the log scale, the skew present in the original population variable remains more

substantial than it represents. The original population variable was then removed, and the log-transformed version was retained.

Principal Component Analysis

Principal Component Analysis (PCA) was used to address the strong correlation among the eight Google Trends variables. Since there was considered overlap amongst the eight variables, PCA was used to reduce their shared information into a smaller number of components. PCA was exclusively fit on the training set. The scaler was fit using only using training data and then PCA was fit on the scaled training data. The fitted transformation then applied to the test set. No information from the test set was used to for the PCA components which was done with intent to avoid any data leakage.

The first three components explained approximately 90.2% of the total variation in the eight search variables:

- **PC1:** 69.5%
- **PC2:** 11.4%
- **PC3:** 9.3%

Retaining three components therefore reduced the number of highly correlated predictors while preserving most of the information contained in the original search variables.

Additional Feature Engineering

Two additional predictors were created based on feedback attained during the first EDA:

- `admission_rate_lag2` — psychiatric admission rate from two years earlier.
- `neighbor_admission_rate_lag1` — previous-year average psychiatric admission rate among neighboring states.

These variables were kept separate from PCA because they represent historical and geographic information rather than Google Trends information. For Alaska and Hawaii, which do not have bordering states, the previous-year national average admission rate was used as the geographic comparison value. That does bring in a limitation of Alaska and Hawaii resembling their own average along this dimension despite both states differing from the rest of the country pretty significantly.

The `State` variable was not encoded during the common preprocessing stage because the appropriate treatment depends on the modeling algorithm. The linear model for the baseline model requires dummy encoding with one category dropped because it implements fixed effects and including every category without dropping one would cause the model to fail. The tree-based models used after the baseline split on features and do not require the same treatment.

Preprocessing Summary

Preprocessing was used to prevent information leakage, make variables comparable, reduce redundant Google Trends information, address skewed population data, and add meaningful historical and geographic predictors. PCA reduced eight correlated Google Trends variables to three components while retaining approximately 90.2% of their information.

Models

Baseline Model

Linear regression was selected as the baseline model because it provides a good benchmark for evaluating more advanced predictive models. The model was built to predict psychiatric admission rates using admission-rate lag features, neighbor admission lag, Google Trends principal components (PC1–PC3), log state population, and state fixed effects.

The data were divided chronologically into 227 training and 133 testing observations. The year variable was removed to reduce potential temporal leakage. A five-fold expanding-window time-series cross-validation was used so that each testing period represented a future time relative to the related training data.

Regression diagnostics included residual plots, a Q-Q plot, and the Durbin-Watson statistic to evaluate linearity, residual behavior, and potential autocorrelation.

Baseline Results

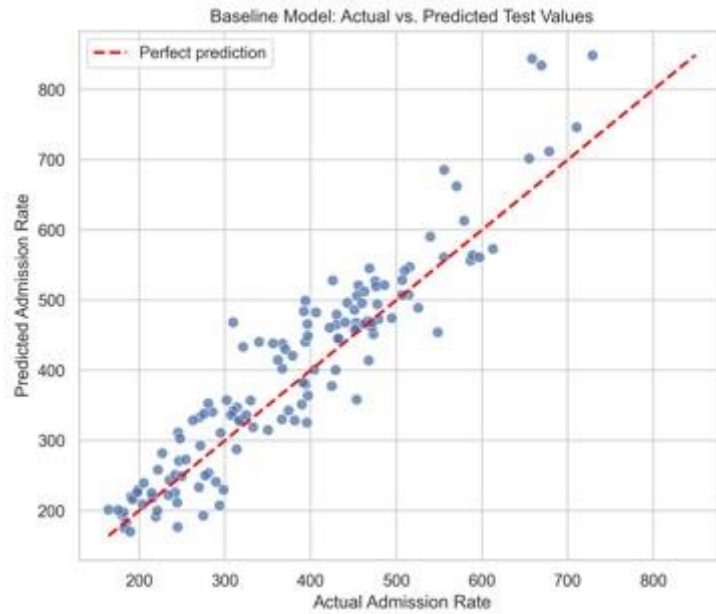
<u>Dataset</u>	<u>RMSE</u>	<u>MAE</u>	<u>R²</u>
Training	38.04	23.61	0.936
Testing	53.98	41.97	0.829

The baseline model explained approximately 83% of the variation in psychiatric admission rates on unseen test data. The time-series cross-validation produced a mean RMSE of 72.88 (SD = 52.03), indicating that prediction error varied significantly across time periods. The first fold performed substantially worse than the others, necessitating further investigation of the first forecasting period for potential causal factors.

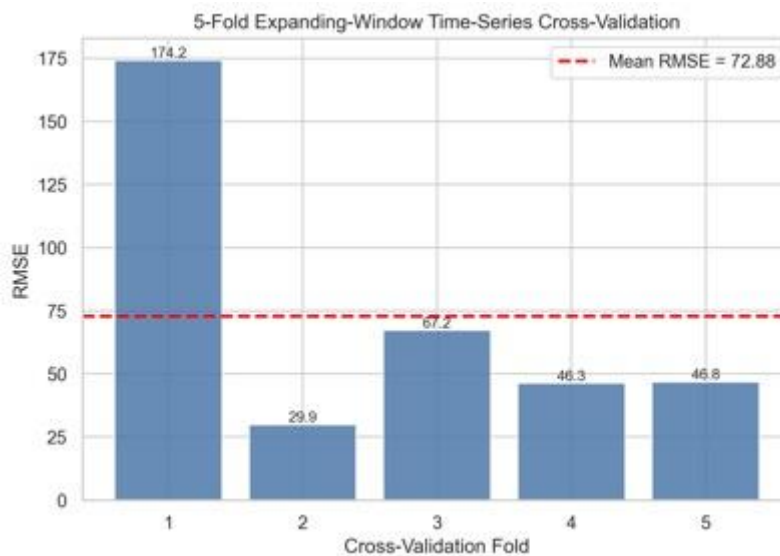
The Durbin-Watson statistic was 1.77, reasonably close to 2, providing weak evidence of strong residual autocorrelation. This alone does not establish independence of error across the state year data as the data spans numerous states and years, and the order of observations is a factor. Likewise, state fixed effects account for differences in average admission rates but don't prove residual independence.

The Q-Q plot tests whether the prediction errors (residuals) fall within a normal distribution. The plot deviates from the line (see figures below) in several places, indicating that prediction errors may be present. The residual vs. fitted plot shows negative residuals at higher predicted admission rates suggesting that the model may be overpredicting at this range. The spread of the residuals seems to change as the predicted values increase suggesting possible heteroscedasticity (amount of error in a model changes depending on prediction levels). Together, the plots below provide warnings that don't fully reject the model but show that caution and further investigation is warranted before strong prediction claims can be made.

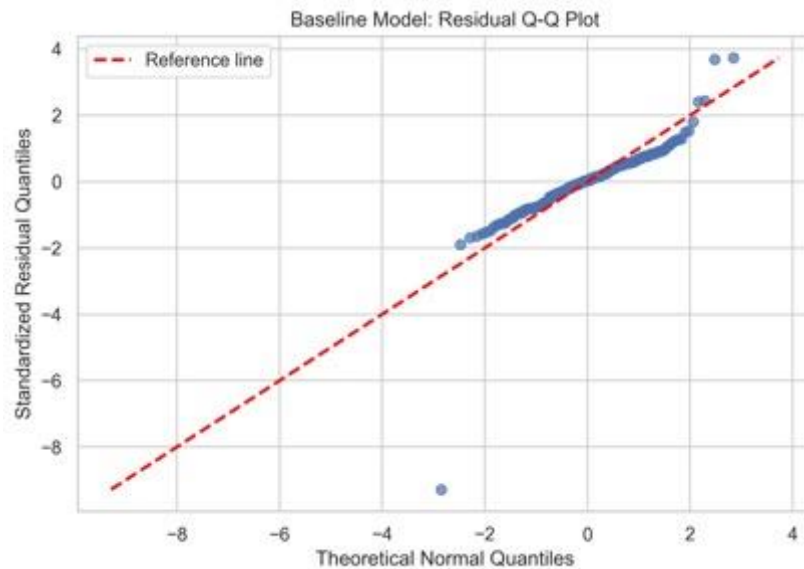
Baseline Figures



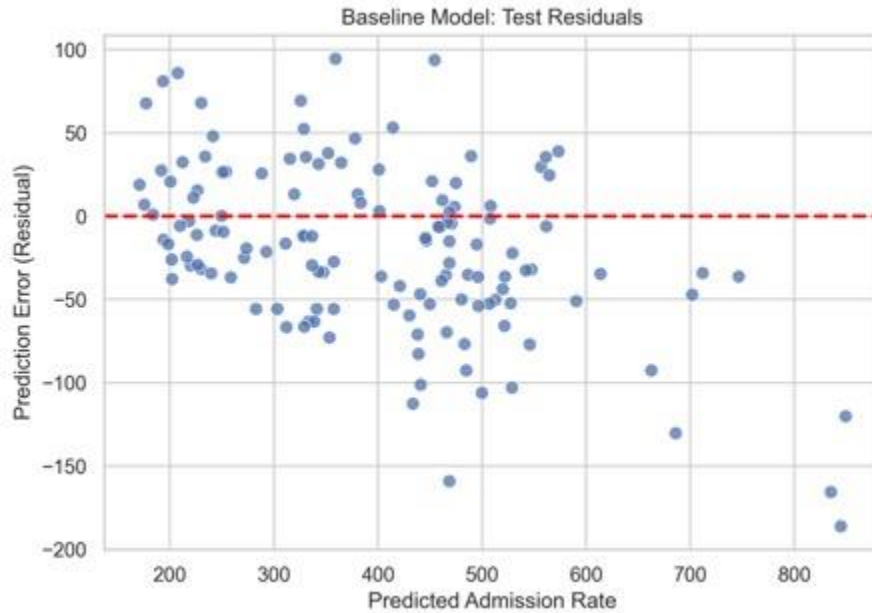
Baseline Figure 1 presents the Baseline Model: Actual vs. Predicted Test Values. Image indicates reasonable predictive accuracy on unseen data.



Baseline Figure 2 presents the 5-fold Expanding Window Time-Series Cross-Validation. Each bar shows the model's prediction error during a different time period – The first fold showed significant prediction error that requires future investigation.



Baseline Figure 3 presents the Residual Q-Q Plot. The plot shows how closely the model's errors follow a normal distribution. Points deviating from the reference line may indicate prediction error.



Baseline Figure 4 presents the Baseline Model: Test Residuals. The plot shows prediction errors on unseen data. The negative residuals at higher prediction values suggests potential overprediction at the upper range; the changing spread is suggestive of heteroscedasticity.

Baseline Summary

The baseline model showed reasonably strong predictive performance and provided a good benchmark for more advanced models. It should be noted, that since cross validation and residual diagnostics showed less than optimum performance, the baseline should be viewed as a benchmark that may require further investigation/refinement. Given project time constraints, such potential improvements will be considered in the ‘next steps’ section below. Since Google Trends variables were included in this model, the results cannot determine whether Google Trends improves predictions beyond historical admission, demographic, and geographic

information. To do this, the final analysis will compare at least one model with and without Google Trends variables.

Supervised Summary

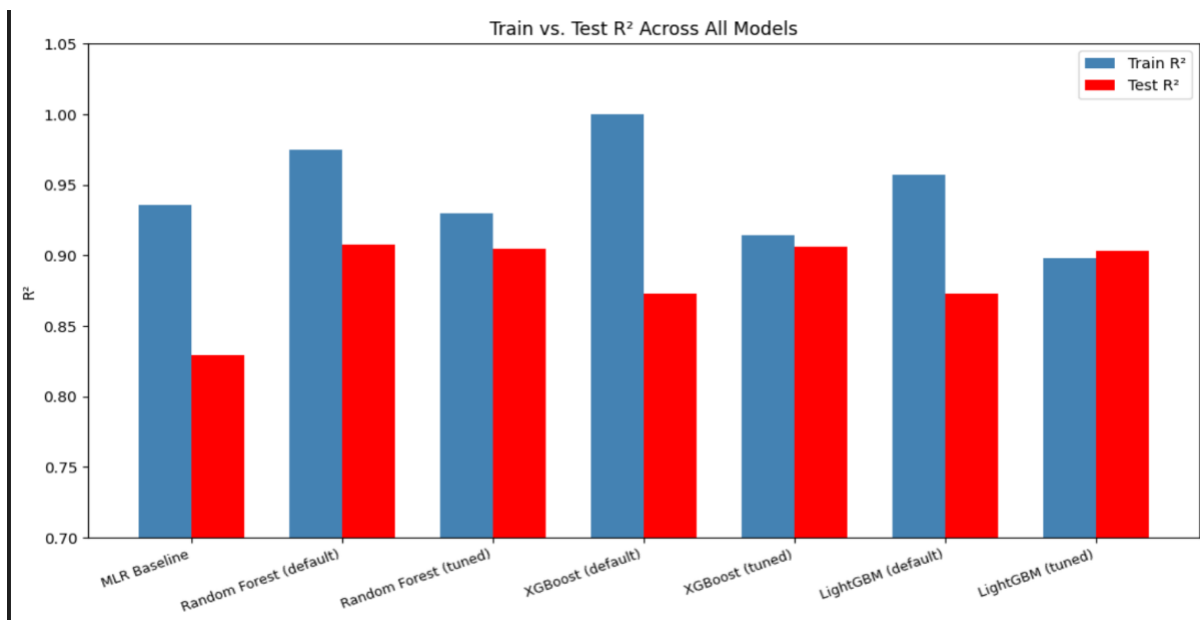
The supervised models that were used in this project were, Random Forest, XGBoost and LightGBM. The Random Forest Model and the XGBoost models were chosen from the original proposal because both models do well at capturing nonlinear relationships and interactions that the linear regression baseline model cannot. The LightGBM model was recommended to be added for a gradient boosting comparison; which would compare the XGBoost model to the LightGBM.

There was a difference in the preprocessing pipeline for the supervised models compared to the baseline model. The baseline model used dummy encoding due to its fixed effects. Whereas these models used the label encoding since the tree-based models do not require the same treatment as the linear model does. The extra dummy encoded columns would make the model larger than it needed to be so the encoding was handled separately for the linear and tree-based models.

The models were fit on the same target, the same chronological split and the same predictors as the baseline model to ensure reproducibility. The predictors that were used across the board were the 3 principal component analysis components, the log population, the admission_rate_lag2 and the neighbor_admission_rate_lag1. Each model started with default settings and evaluated with a 5-fold expanding window time series which again was the same as the baseline to make sure comparison was done fair and correct.

Tree based models do not carry the same assumptions as the linear regression model so there is no linearity or residual normality/independence since the tree-based models do not assume that the data will fit a specific shape or curve and find patterns and jumps within the data without the need for a straight line.

Initially the supervised models resulted with overfitting which can be seen in figure 5. The XGBoost model had severe overfitting with a train R squared of 1.00 and a test R squared of 0.873 which is a significant gap indicating the model just memorized the data. The hyperparameter tuning was necessary because of this overfitting and hoped to reduce overfitting and improve model performance.



Supervised Figure 5. Train vs test R squared for all the models, showing XGBoost default with the largest gap.

Tuning & Validation

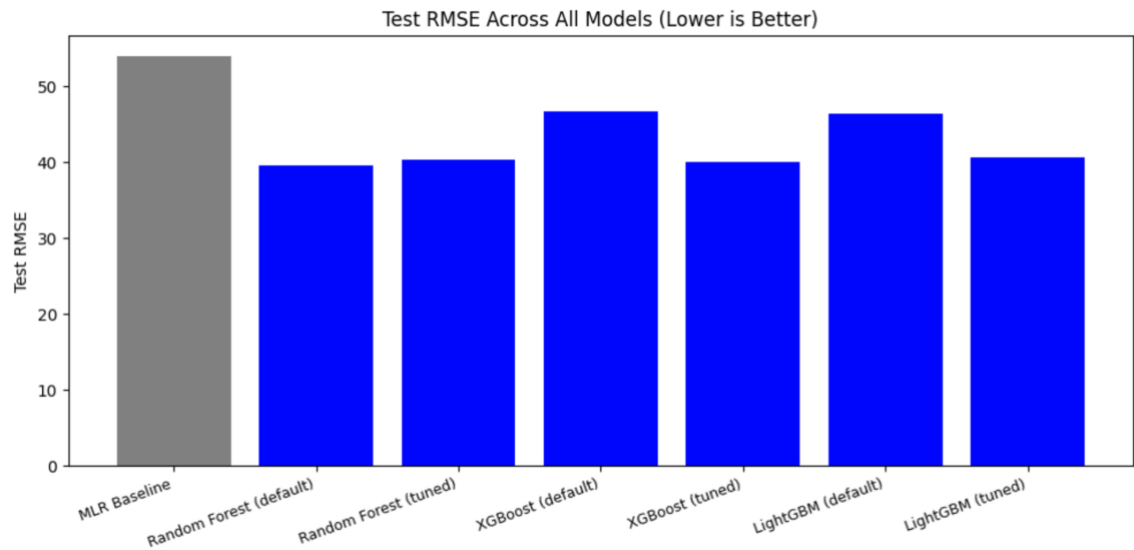
The hyperparameter tuning was done using GridSearchCV which was able to test different combinations of hyperparameter values and reports on the different combinations that perform best. So instead of evaluating each combination with one train/test split the GridSearchCV used the same expanding window time series cross validation that was applied throughout this process. This ensured that all the combinations that were tested were evaluated and validated the same way the default models were. It trained only on the earlier folds and then tested on the later ones which helped avoid leakage during the tuning.

Each of the different default tree-based models were tuned on different parameters that controlled the model's complexity. The reason for this was because of the overfitting seen in the default models and overfitting can be a result of the model being too complex and learning the training data perfectly. The models did not show signs of weak capabilities; the core problem was the overfitting. The Random Forest model was tuned on several hyperparameters so the `max_depth`, `min_samples_leaf`, `max_features`, and `n_estimators` were altered to control how large, and which specific trees can grow. Tree ensembles can overfit due to some of the individual trees becoming too large and memorizing the training data so controlling those helps reduce that overfitting. For the XGBoost, `reg_alpha` and `reg_lambda` were added since they help penalize model complexity. Also, `n_estimators`, `max_depth`, and `learning_rate` were tuned. Clearly with the overfitting the model's complexity was too much and fully memorizing the training set. So, this tuning tried to reduce that issue. For the Light GBM model, `min_child_samples` was added on top of the usual depth controls which controls the minimum data needed before a split is allowed. This was done because LightGBM was struggling with the smaller training data provided.

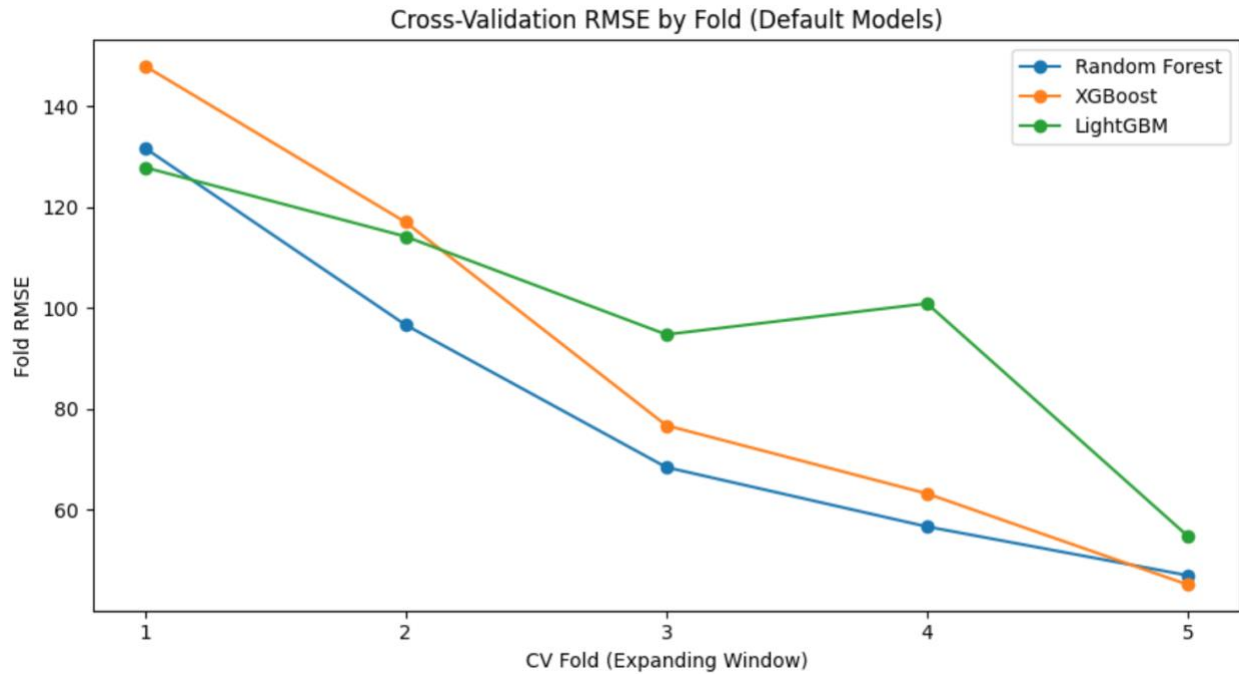
Models' performance for all of the models made including baseline, default tree-based models and tuned tree-based models were evaluated and compared directly, and the results can be seen in Table 1, which contains all the metrics used to evaluate each model. The tuning process was successful and reduced the train test R squared gap significantly for both XGBoost and Random Forest as those two models showed the highest sign of overfitting in the default models. The XGBoost gap dropped from 0.127 to 0.008, and the Random Forest gap fell from 0.067 to 0.025. The best cross validation RMSE for each model was 76.83 for Random Forest, 84.99 for XGBoost and 87.83 for the LightGBM. Figure 6 shows the test RMSE across all the different models and Figure 7 shows cross validation RMSE by fold for the default models. This figure should be interpreted carefully, in an expanding window design each successive fold trains on more data and evaluates a different future period so the declining RMSE could reflect the model improving with more training data, or later years being easier to predict or a mix of both. The two explanations cannot be found by the fold RMSE alone. Along with that the cross validation RMSE and test RMSE do not explain a consistent story which does not mean there is an error, but a later period could be easier to predict than earlier validation periods. Figure 8 shows the train/test R squared gap for every model for default and then for the tuned models. These figures indicate that the tuning improved generalization for every tree-based model and the tuning improved test performance for XGBoost and LightGBM models.

Model	Train RMSE	Test RMSE	Train R ²	Test R ²
MLR Baseline	38.04	53.98	0.936	0.829
Random Forest (default)	23.88	39.59	0.975	0.908
Random Forest (tuned)	39.73	40.31	0.93	0.905
XGBoost (default)	0.06	46.66	1.0	0.873
XGBoost (tuned)	44.04	40.04	0.914	0.906
LightGBM (default)	31.09	46.49	0.957	0.873
LightGBM (tuned)	48.15	40.64	0.898	0.903

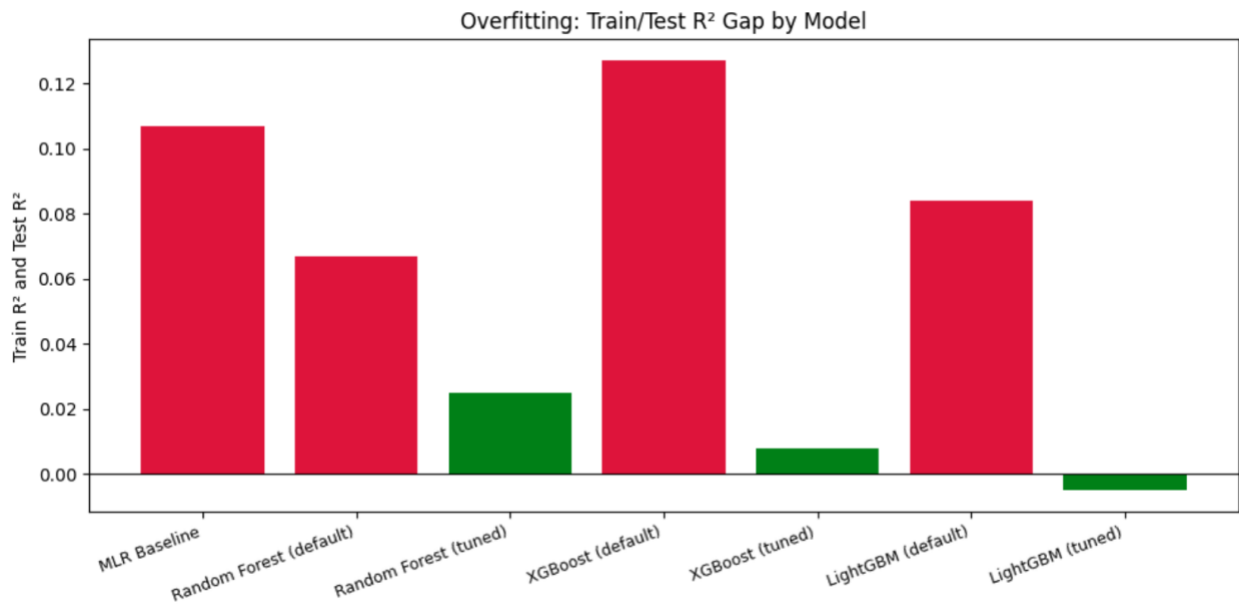
Supervised Table 1. Results for all models created including baseline



Supervised Figure 6. The train vs Test RMSE results for all different models including baseline, default and tuned models



Supervised Figure 7. Cross validation RMSE for all default models showing a steady decrease.



Supervised Figure 8. Train/Test R squared gap for all models created

Final Model

The final and best model that was created would be the tuned XGBoost model, though it should be mentioned that the differences between the top performing models are small enough that there is effectively a tie in raw predictive performance rather than an obvious winner. The tuned Random Forest model had nearly the exact same R squared value as the XGBoost model but the XGBoost model outperformed the Random Forest model in the train/test gap with a value of 0.008 vs Random Forest value of 0.025. This means that the XGBoost model generalizes more reliably than the Random Forest and performs pretty much the exact same. The XGBoost model also performed better than the LightGBM model with a better R squared and the LightGBM model showed the most fold to fold CV instability. The XGBoost model benefitted a lot from tuning as it had much better performance and nearly no more overfitting after tuning. Random Forest on the other hand improved generalization but did not improve as much in test performance. All the tuning models outperformed the baseline model as well.

Figure 9 shows the actual vs predicted admission rates for XGBoost on the test set of data and shows the points clustered closely around the perfect prediction line through the middle of the graph. It can be seen that the model's errors are not biased towards over or under predicting at any of the admission levels so the model is able to perform well with high or low admission rates. Higher values of `admission_rate_lag_2` shown in red push predictions toward higher admission rates, while lower values shown in blue push predictions lower, confirming the expected positive relationship between historical and predicted admission rates.

Figure 10 shows the feature importance for the Random Forest and the XGBoost models and shows that the `admission_rate_lag2` is the dominant predictor in the model's performance. This

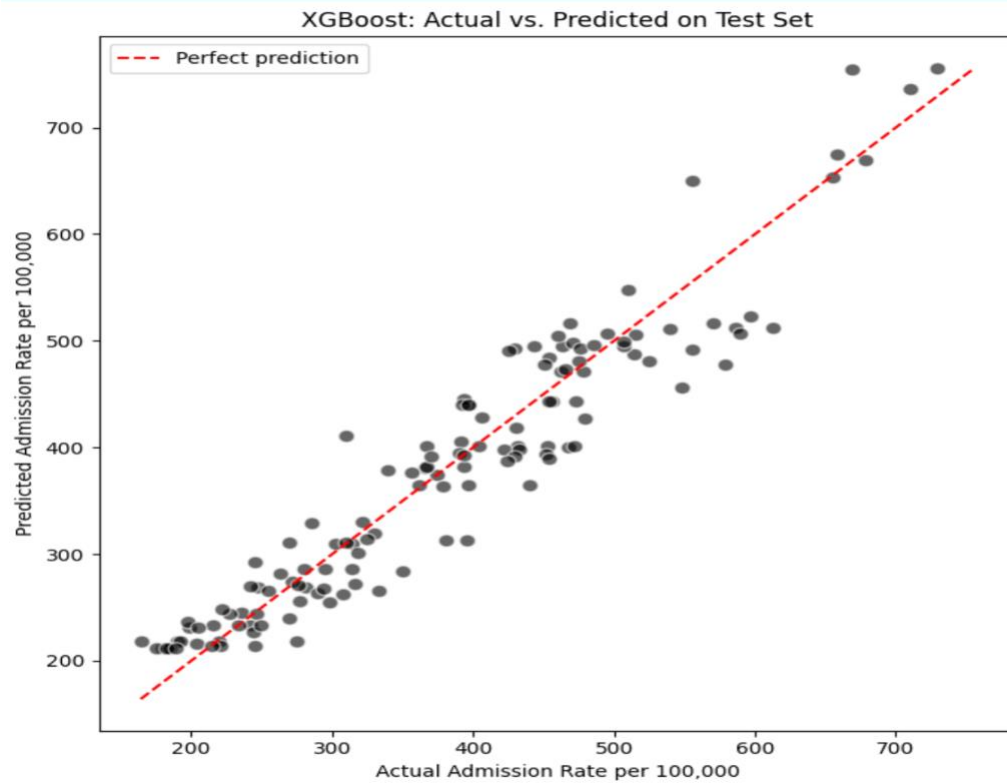
agreement across the two structurally different algorithms suggests that this finding reflects a pattern in the data rather than an artifact specific to how one of the algorithms captures performance. Figure 11 shows the SHAP summary for the XGBoost which adds to the directionality to the feature importance finding.

An ablation study was conducted to directly test the incremental predictive value of the Google Trends features created during PCA. Random Forest and XGBoost were each refitted using the same exact hyperparameters once with the 7 features that included the PCA features and once without the PCA features. The study showed that both models performed better without the Google Trends predictors than the model using the PCA Google Trends predictors. This shows a stronger finding than no difference as it indicates that with this model architecture and the data being used Google Trends search interest data does not provide meaningful predictive value beyond what the historical admission patterns, spatial information and the population provide. This analysis indicates an answer to the research question of how well psychiatric admission rates can be predicted and what indicators are most useful for that prediction. Admission rates can be predicted well, the XGBoost model achieved a test R squared of 0.906 meaning it explains around 91% of the variation in future psychiatric admission rates. The indicators that drive the model's performance though are not the Google Trends search interest data but historical admission patterns and spatial information from neighboring states. This does not support the original hypothesis proposed.

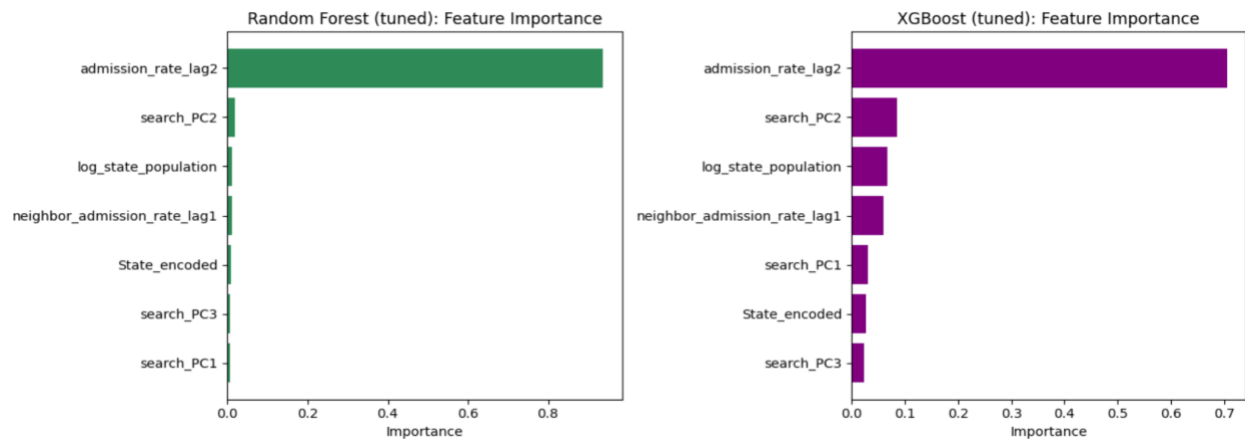
This does not mean that the project has no value but shows that if there are stakeholders focused on anticipating near time psychiatric resource needs the takeaway should be that the state's own historical admission trajectory and the trends from neighboring states provides the best features for forecasting.

It is worth noting a limitation of the State variable and the encoding that was done for it in these specific models. Label encoding assigns an arbitrary number that is used as a code for each state which imposes an ordinal relationship that does not reflect anything meaningful. There would be no reason for one state to be a lower number than another state. Tree based models can still find useful splits within this encoding, but a more appropriate approach would be target encoding as it would represent each state using information actually related to the outcome. Another option could be geography-based representation.

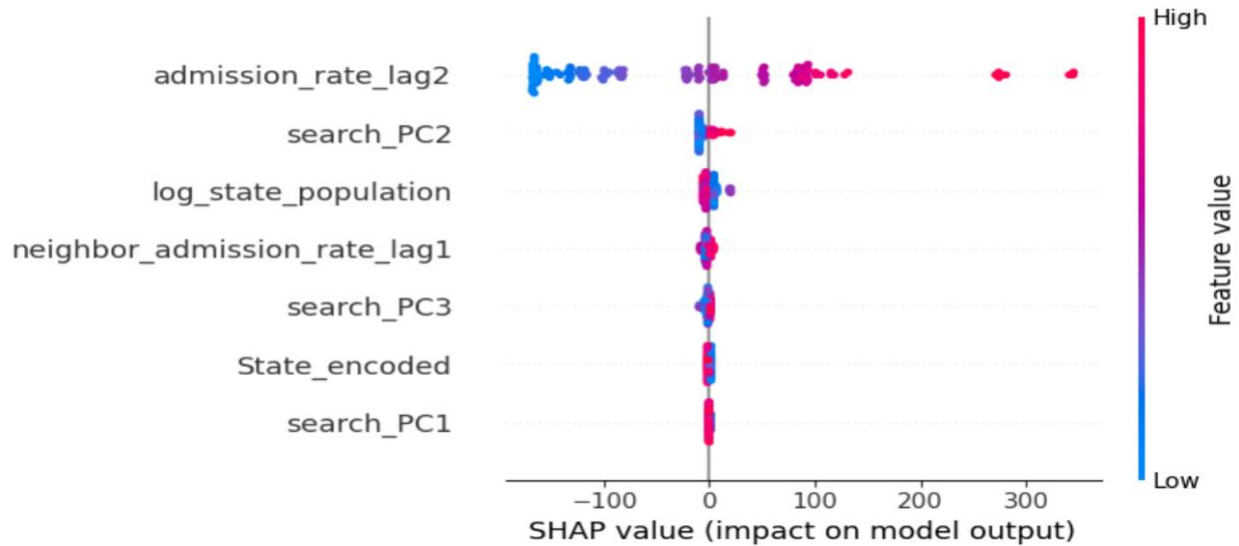
It is important to state that the Google Trends search interest data is not necessarily unable to provide any meaningful predictive power but that the Google Trends search terms when combined into three principal components and evaluated at a one-year lag did not improve predictive power more beyond historical admission patterns and neighboring states trends, population and state identity using this architecture and algorithms. Choosing different Google Trends search terms or a different lag structure or a different way of combining them could carry incremental value that was not seen here. Which also connects to the multicollinearity seen with the original 8 search terms during preprocessing where a single principal component captured around 70% of all the shared variance. This may indicate that the search terms captured one mixed signal from all the search interest rather than 8 different search interest information.



Final Model Figure 9. XGBoost Actual vs. Predicted scatter plot



Final Model Figure 10. Feature Importance for Random Forest and XGBoost models



Final Model Figure 11. SHAP value impact for XGBoost

Project Specific Feedback

Originally in the project proposal, Team Rho was given feedback based on the sample size of the data the team was using. It was pointed out that the sample size was small which might lead to overfitting. The team continued with the data it had originally found and did run into some overfitting from default tree-based models. But during hyperparameter tuning, the overfitting was reduced and the models were able to generalize well.

During the exploratory data analysis, the project manager for team Rho advised that feature engineering be prioritized over machine learning algorithms. Specifically, the temporal and spatial structure already present in the data might be used to add temporal and spatial lag predictors that may potentially add depth to the final analysis. It was also suggested that state level predictors such as income, insurance, and admissions capacity be added for a more thorough analysis.

The group was also suggested to either group terms into conceptual categories such as suicidal ideation, self-harm and help-seeking to use PCA as a data driven alternative. Team Rho decided to use PCA instead of clustering terms. The PCA strategy loosely followed this suggested structure anyways as PC3 separated personal crisis expression from help seeking language.

Project time constraints forestalled the later advice, but in keeping with the former, team Rho chose to add two new predictors in line with manager advice. These included ‘admission_rate_lag2’, a psychiatric admission rate from two years earlier, and ‘neighbor_admission_rate_lag1’, the previous-year average psychiatric admission rate among neighboring states. The addition of these variables proved efficacious, as both of the new predictors demonstrated significant feature importance during the hyperparameter stage of model development (see tuning section).

Discussion & Next Steps

Key Takeaways

As indicated in the final analysis, within the context of this study, Google Trends data did not prove to be a reliable predictor of future inpatient admissions. The results indicate that historical & geographic patterns prove to be much stronger indicators of future inpatient admissions. The baseline model explained roughly 83% of variation in unseen test data. Cross-validation and residual diagnostics showed significant variability in prediction error across folds and possible heteroscedasticity. Advanced modeling demonstrated the strong predictive utility of historical

admissions with geographic patterns proving to be a lesser but significant predictor as well.

Within the limitations of the study, the hypothesis was not supported and google trends data alone did not demonstrate significant future predictive value.

Based on the analysis, the historical admissions patterns and neighboring-state trends are strong predictors of future psychiatric admission rates. This finding is supported by feature importance SHAP analysis and the ablation study performed.

We cannot state that Google Trends search interest data has no predictive value in general but that the 8 search terms that were used in this project summarized into 3 principal components at a one year lag did not add value to the other predictors in the model architecture that was used.

Limitations

In addition to the cross-validation and residual problems noted within the baseline model, the study as it currently stands does present several limitations that may stifle conclusive hypothesis testing. For example, the Google Trends measurement of relative versus absolute search interest, the use of PCA's rather than a comprehensive set of individual suicide related search terms, and the HCUP measure of observed utilization rather than actual demand are only a few of the factors that limit hypothesis testing. State to state differences in access, standards, and insurance coverage may also affect the observed admission rates.

Next Steps

For a more comprehensive analysis of the original hypothesis regarding the potential predictive value of google trends data, several modifications could be made for improved future hypothesis testing:

- Add state-level income, insurance coverage, psychiatric bed capacity, and workforce measures. Such data may add predictive value over the present study.
- Investigate the high error observed in the first cross-validation period to determine what factors in the data-generating process contributed to poor performance.
- Consider testing a more comprehensive suicide search term set instead of PCA predictors to see if specific terms contain predictive value obscured by PCA predictors.
- Test whether additional temporal and spatial lags add predictive value
- Compare more models with and without google predictors to test for small predictive value.

Some issues to address before the model can be used for real world resource allocation and future admission predictions are:

- Switching the state encoding from label encoding to geographical or target encoding to ensure there is no arbitrary ordinal structure on a categorical variable
- The cross validation instability should be addressed as the fold level performance varies significantly and it has not been found whether this is due to genuinely harder to predict earlier periods or if there is a lack of training data for early folds or the presence of outliers or a mix of it all.

- The Google Trends search terms could be looked into further or switched with others.

The specific findings from this project should not generalize Google Trends search interest data as a whole. A different set of terms or a different architecture could result in a different outcome.

The addition of such improvements would provide future data analysts with a more complete answer to the project question:

‘How well can state-level psychiatric admission rates be predicted using suicide- and crisis-related Google Trends data, geographic information, and historical admission patterns?’

Code

The code, notebooks and datasets can be accessed here: <https://github.com/brendancollari/Team-Rho-Capstone-Project>

The coding for this project was conducted using Python in Jupyter Lab. There were many different python packages used within the notebooks; the packages were pandas and NumPy for the data manipulation. Scikit-learn for preprocessing, specifically StandardScaler and LabelEncoder, PCA was used for dimensionality reduction. The RandomForestRegressor built the Random Forest model. TimeSeriesSplit and GridSearchCV did the cross validation and tuning. The mean_square_error and r2_score calculated the metrics. XGBoost and LightGBM were used for gradient boosting models and SHAP was used for more model interpretation.

Matplotlib was used frequently for visualization and a random state of 42 was used as a random seed.

The repository in GitHub can be used to find the data used in the project which is organized into data/raw and data/processed folders. The raw folder has the original data that was found which includes the Google Trends data, the HCUP data and the U.S. Census Bureau data. The processed folder contains cleaned and feature engineered data including the final train and test split data used for the models created. That data is within the Engineered_Data_v2 folder of the processed folder. The repository also contains a Notebooks/ folder that contains a notebooks/raw and Notebooks/processed folder as well. The processed notebook folder contains all notebooks used and have relative paths so the notebooks can be replicated. A README.md is also in the repository that describes the project and has the repository structure within.

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Appendix

Data Dictionary

Variable	Description	Type	Units	Source	Role
State	U.S. state	Categorical	N/A	Census	Identifier
Year	Observation year	Integer	Year	Census	Identifier
log_state_population	Log transformed state population	Numeric	Log Persons	U.S. Census Bureau	Predictor/Control
mental_health_admissions	Annual inpatient psychiatric admissions	Numeric	Admissions	HCUP	Supporting variable
admission_rate_per_100k	Psychiatric admissions per 100,000 residents	Numeric	Rate per 100,000	HCUP + Census	Outcome (Target)
suicidal_thoughts_lag1	Previous year's Google Trends interest for "suicidal thoughts"	Numeric	Google Trends index (0–100)	Google Trends	Source variable (turned into PC1-PC3)
suicide_hotline_lag1	Previous year's Google Trends interest for "suicide hotline"	Numeric	Google Trends index (0–100)	Google Trends	Source variable (turned into PC1-PC3)
self_harm_lag1	Previous year's Google Trends interest for "self harm"	Numeric	Google Trends index (0–100)	Google Trends	Source variable (turned into PC1-PC3)
mental_health_crisis_lag1	Previous year's Google Trends interest for "mental health crisis"	Numeric	Google Trends index (0–100)	Google Trends	Source variable (turned into PC1-PC3)
suicide_prevention_lag1	Previous year's Google Trends interest for "suicide prevention"	Numeric	Google Trends index (0–100)	Google Trends	Source variable (turned into PC1-PC3)
crisis_hotline_lag1	Previous year's Google Trends interest for "crisis hotline"	Numeric	Google Trends index (0–100)	Google Trends	Source variable (turned into PC1-PC3)

psychiatric hospital_lag1	Previous year's Google Trends interest for "psychiatric hospital"	Numeric	Google Trends index (0–100)	Google Trends	Source variable (turned into PC1-PC3)
depression_help_lag1	Previous year's Google Trends interest for "depression help"	Numeric	Google Trends index (0–100)	Google Trends	Source variable (turned into PC1-PC3)
search_PC1	First principal component of the 8 Google Search terms, general crisis search intensity	Numeric	Compon ent Score	Derived	Predictor
search_PC2	Second principal component, contrasts suicide/drpressi on help seeking vs crisis language	Numeric	Compon ent Score	Derived	Predictor
search_PC3	Third principal component, contrasts personal crisis vs help seeking language	Numeric	Compon ent Score	Derived	Predictor
admission_rate_lag 2	Two year lagged psychiatric admission rate per 100,00	Numeric	Rate per 100,00	HCUP and Census (derived)	Predictor

neighbor_admission_rate_lag1	Prior year average admission rate of state's bordering states	Numeric	Rate per 100,00	HCUP and Census (derived)	Predictor
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Descriptive Statistics of Final Analytical Dataset

Variable	Description	Data Type	Non-Missing	Missing	Unique Values	Range Categories	Mean (numeric)
State	U.S. state associated with the observation.	str	360	0	46	46 unique values	
Year	Calendar year of the observation.	int64	360	0	8	2016.0000 to 2023.0000	2019.4861
admission_rate_per_100k	Psychiatric admission rate per 100,000 population; model target.	float64	360	0	360	110.107 to 826.9086	394.4748

admission_rate_lag2	Psychiatric admission rate from two years earlier.	float 64	3 6 0	0	3 6 0	110.1 107 to 826.9 086	409.2 245
neighbor_admission_rate_lag1	Previous-year psychiatric admission rate among neighboring states.	float 64	3 6 0	0	3 4 5	190.4 005 to 640.7 832	405.6 167
search_PC1	First principal component summarizing Google Trends search activity.	float 64	3 6 0	0	3 6 0	- 4.974 8 to 4.210 8	0.237 9
search_PC2	Second principal component summarizing Google Trends search activity.	float 64	3 6 0	0	3 6 0	- 2.807 0 to 2.188 1	- 0.243 2

search_PC3	Third principal component summarizing Google Trends search activity.	float 64	3 6 0	0	3 6 0	- 2.613 1 to 2.517 8	- 0.356 3
log_state_population	Natural-log-transformed state population.	float 64	3 6 0	0	3 6 0	13.26 92 to 17.49 02	15.26 48